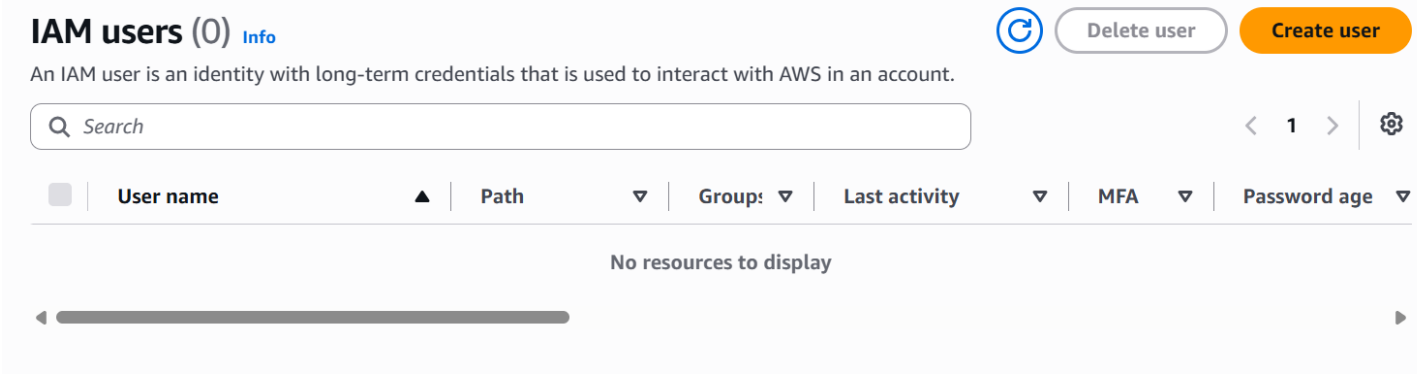
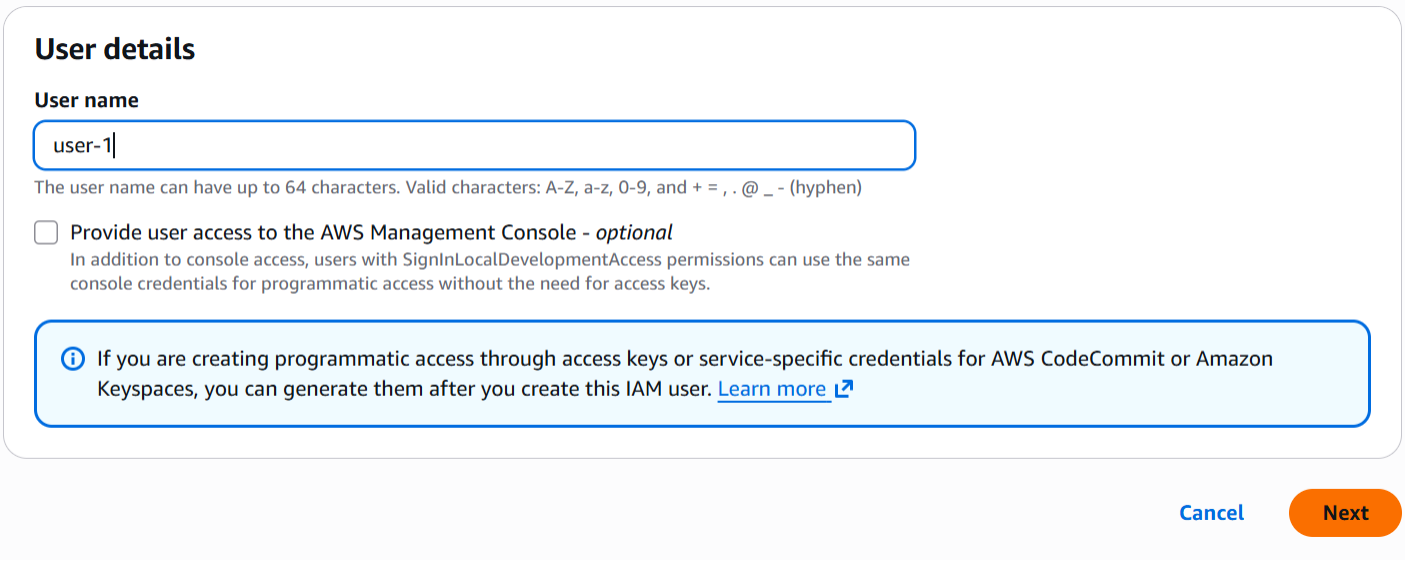
I AM User Task

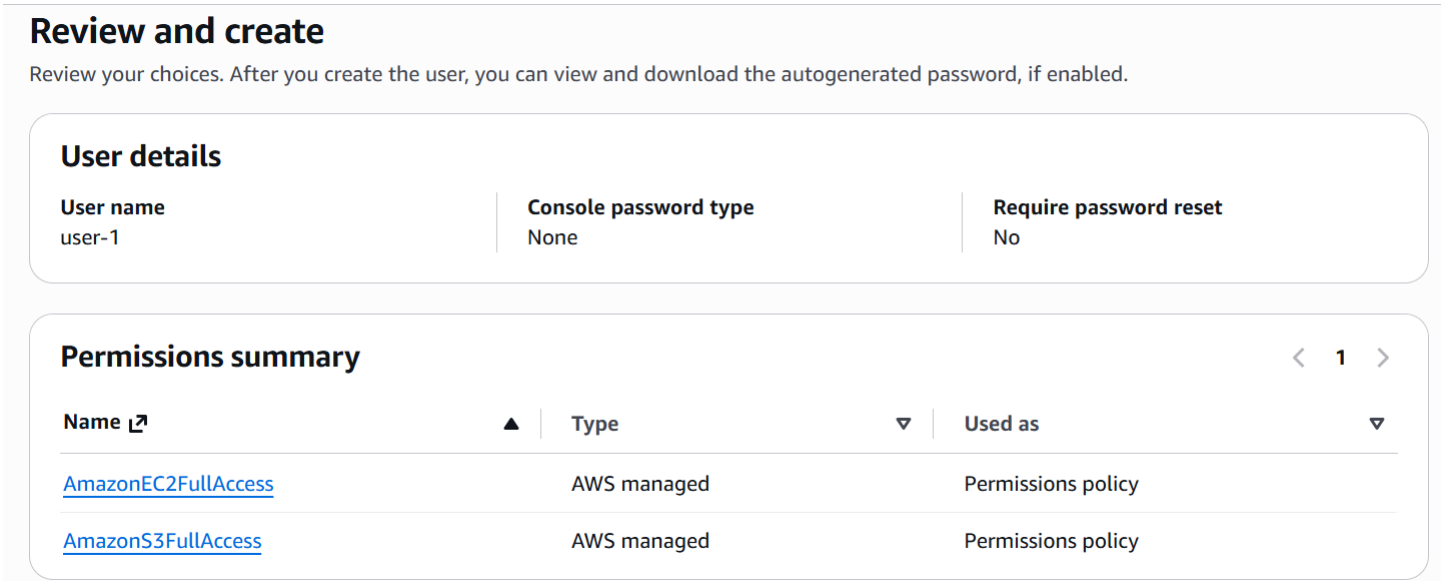
1. Create one IAM user and assign EC2 and S3 full access roles.

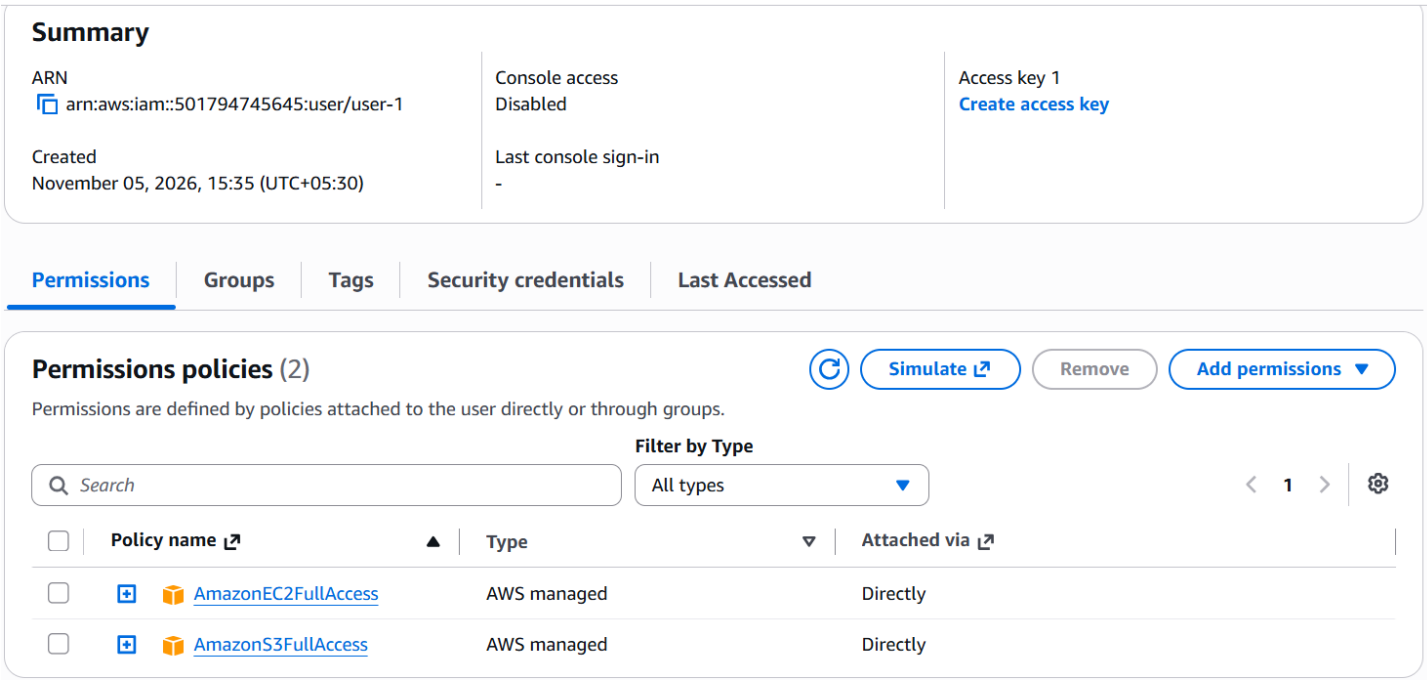
Steps:
Go to IAM user in AWS services → then clicked on Created user → Given User-name
→ Selected permissions - AmazonEC2FullAccess & AmazonS3FullAccess → USER Has Been Created...

Results Below:





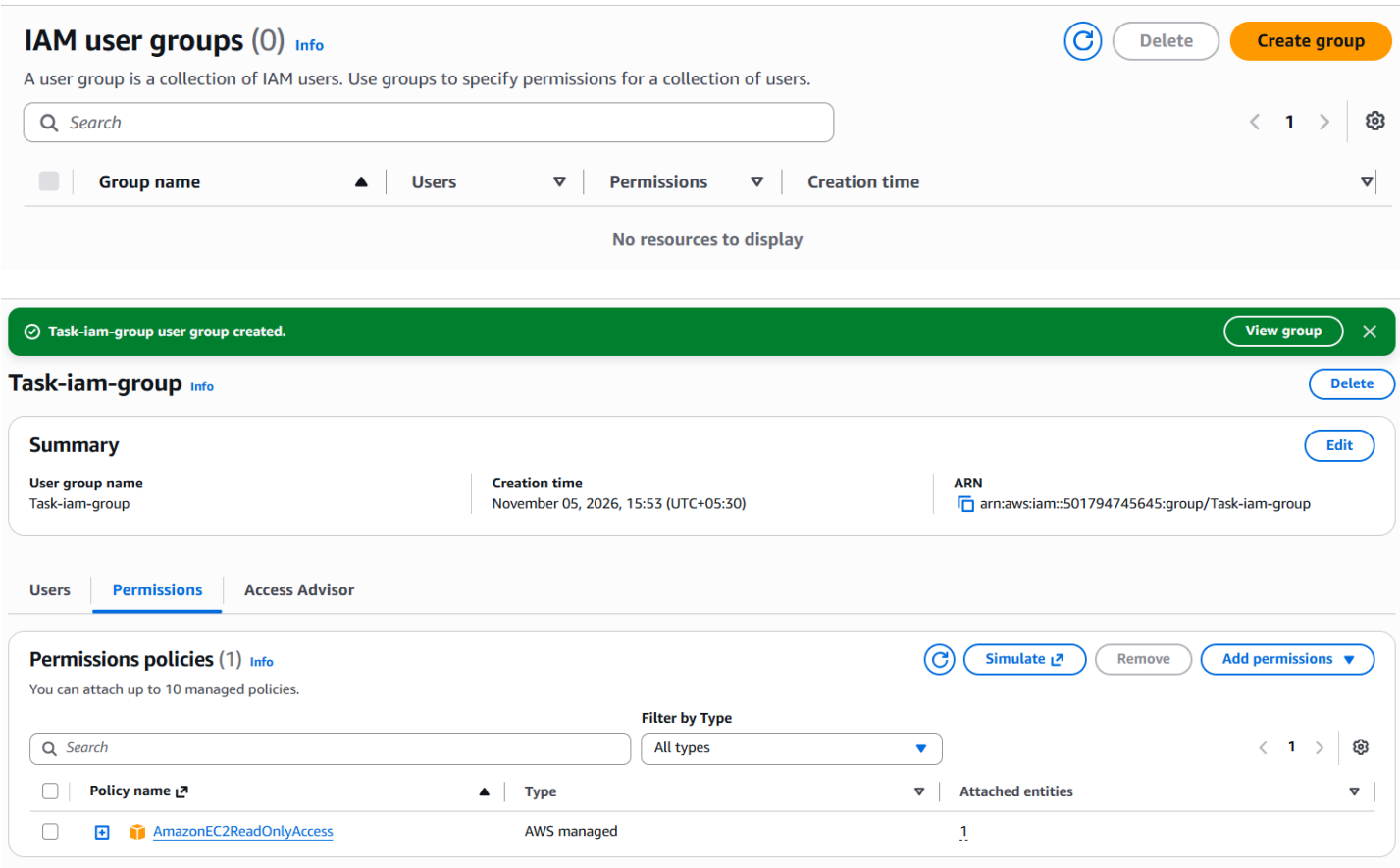




2. Create one group in IAM and assign read access for EC2.

Steps:
Goto IAM User and select IAM User Group → then clicked on Created Group → Given Group-name → Selected permissions - AmazonEC2ReadOnlyAcess → GROUP Has Been Created...

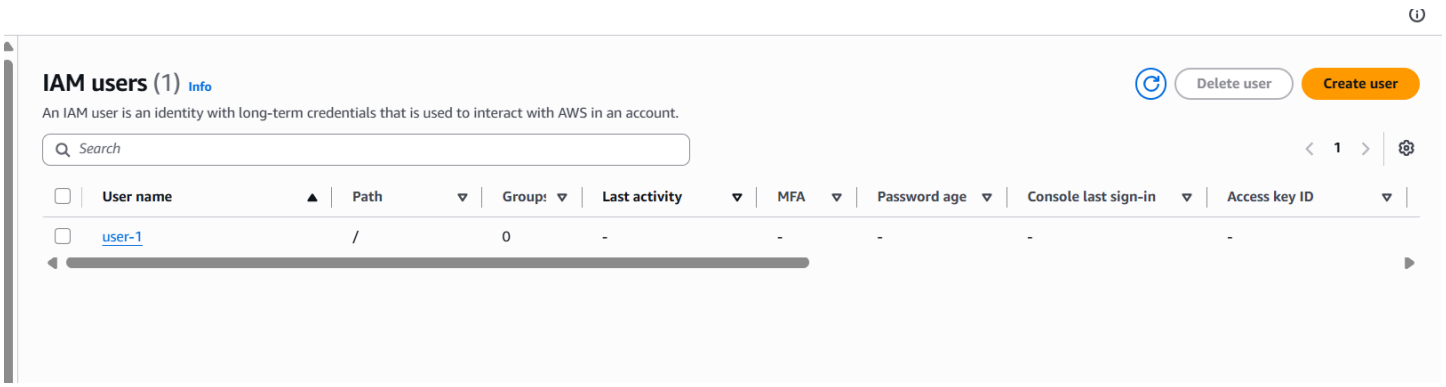
Results Below:

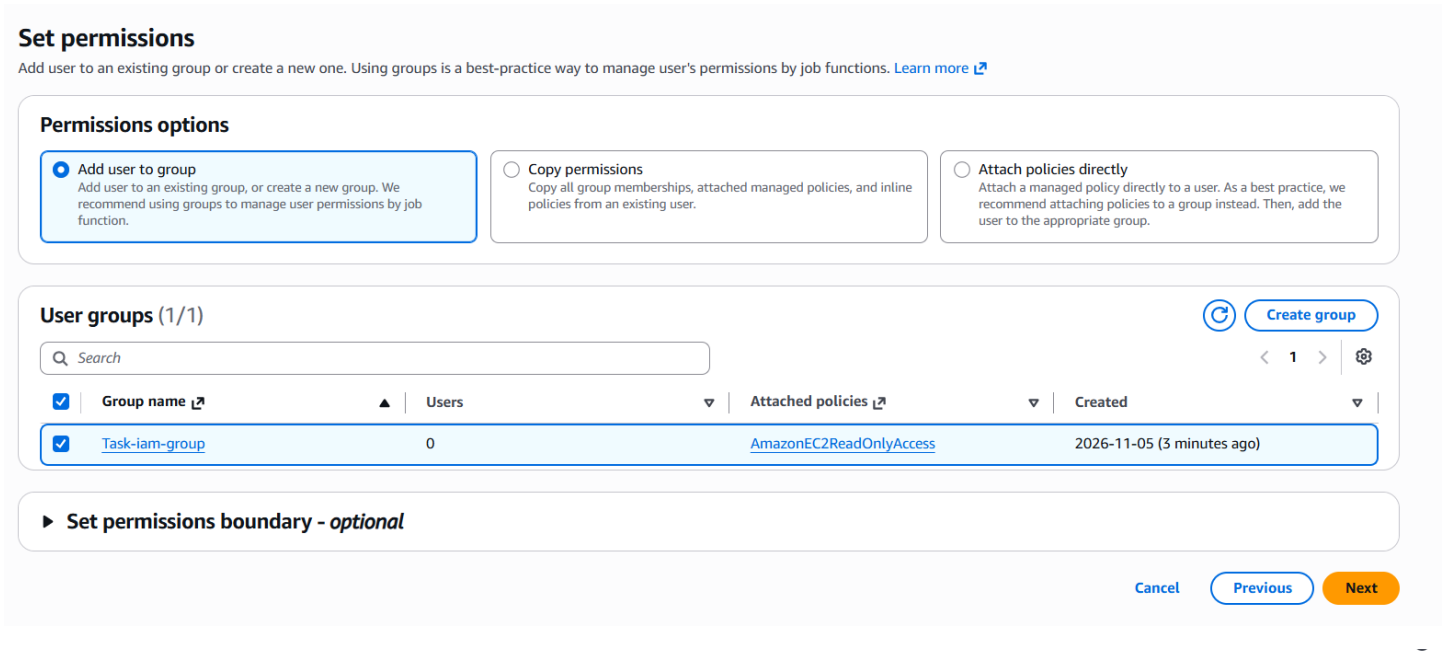


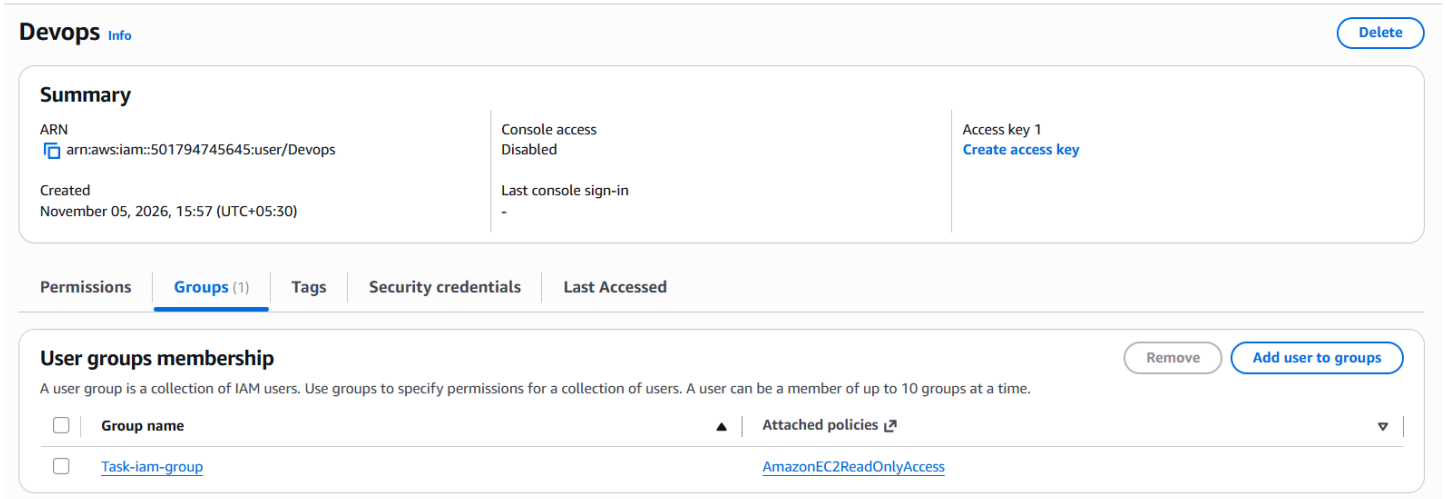
3. Create a new user named "Devops" and add to the group created in task 2.

Steps:
Go to IAM user in AWS services → then clicked on Created user → Given User-name
→ Selected Add User to group → Select the option of Group → USER Has Been Created...

Results Below:







4. Write a bash script to create an IAM user with VPC full access.

Steps:
Login into AWS CLI

```
[ec2-user@ip-172-31-16-171 tmp]$ vi create_vpc_user.sh
[ec2-user@ip-172-31-16-171 tmp]$ cat create_vpc_user.sh
#!/bin/bash

# Exit immediately if a command exits with a non-zero status
set -e

# Configuration
USER_NAME="vpc-admin-user"
POLICY_ARN="arn:aws:iam::aws:policy/AmazonVPCFullAccess"

echo "Creating IAM user: $USER_NAME..."
aws iam create-user --user-name "$USER_NAME"

echo "Attaching AmazonVPCFullAccess policy to $USER_NAME..."
aws iam attach-user-policy \
  --user-name "$USER_NAME" \
  --policy-arn "$POLICY_ARN"

echo "IAM user '$USER_NAME' has been created with full VPC access."
[ec2-user@ip-172-31-16-171 tmp]$
```

```
aws: [ERROR]: Failed to decode the verification code.
[root@ip-172-31-16-171 tmp]# aws login --remote
Browser will not be automatically opened.
Please visit the following URL:

https://us-east-1.signin.aws.amazon.com/v1/authorize?response_type=code&client_id=arn%3Aaws%3Asignin%3A%3A%3Adevtools%2Fcross-device&state=2948592b-44dc-499e-bb65-da039a89
1fa&code_challenge_method=SHA-256&scope=openid&redirect_uri=https%3A%2F%2Fus-east-1.signin.aws.amazon.com%2Fv1%2Fsessions%2Fconfirmation&code_challenge=0w69OWuiyAHI5FDvyjp
NbD60YAw6t6Zegd307ifu1E

Enter the authorization code displayed in your browser: Y29kZT1leUo2YVhBaU9pSkVSVVlpTENKbGJtTWlPaUpCTWpVMlIwTk5JaXdpWVd4bklqb2laR2x5SW4wLi5vUFc3XOlIVE5nLTljUW5BLlZqdU1BeGV
VnJSbzNCVzdKajUxRGRFMHJBdmJZX1FwQ2haai0weXZTM1BZYktPanpjbHZWVHFkdGxiclNQaHItaUh4YOVNZDNZNThtU1Z3VE4lOUpkeE1OQlg1dWY2aHdLZ2ZqR28OZjktN184LTlQNGEtalB1NnJYUDB6cndyNUxLUGJRM1FU
RnZoQlZvdXp2dFRybkRGMU5EWkNWV2hjU2pBa2NBcGpuNVZEbnd2T2x6dEVzTmJWZFVuMFNLRkhYVXdXYWJfc1FoNU5nZTd6anMycHZycWhXSHRsNUtDUzdOeGpHNOJQZ2dKRXBoaS1NcnVndXpkVllSTjJfMUZyaOpaOHpVSFh
UUFjZ1BRSzFTLVVTMko1MlR2UEYzSOFwYkRqMER6dHdjdEZuTHNxTEVTQW5wNWFnNOY1YUh1NFF5UE5IcWhMeXBKVDdpV3FGR2EyNnFOYnp3alBTVnVLemV2cFh0TGxTckoOSUQ4ROlUMXJyb2lqNGJaMnBLcERXWm9aTjNuVXZ
cOZoamQONEkOeUdnVwRGWHR5eE9ZT1VMbURTTTEwdURmdWtyQXpmaDJiajVtWUZQeGJtSTNOQkJZNTM2VFJFbjFZVWE5cWtRT2MxWENFZWhKYm5OMERPLXhMWWhTMmh2Z2otN2Fuc284VW9BUS1EU3h2VFhmRzE2UUU5ckVseGF
b1gtLTZTZkN0WnJES1d2eHUzR1BrYlFpYkdKTURFUUO0Z1VwQy1NOFIyVXJtMnBHVzQyQ1hGVFdSN0lQYkt3WjhIMVZ0Swx1OU5WTFVGR0ljNnN3V2ROWk5YnpYQjloMEVmZ3JFWE5TTmxSRzJKM2tOYUQxQXQyRWZPSTZQZHR
T2xtVVkxQjJ6VTZ3OC1RN3JhZWVGQXFMU1liRldUUUdJeTdyUmFzWURfUl84WVN2TmRiRHQOT256Ml9BOGRvcjRJNC1pVHpFNjZmdHNjNDJWTmpCYmpHdVdSODBuTU5fNTJ6bnNLaDRSaOlpRXZVd3lUaTJlU2NWeUJoOUQyTXp
TkZkWkxqRWtvOWM2a1N2SXhRS3ZORUgwcTJEcC1TX1c2Z3VwSTU2cExmxO9sTkJWbGZrMVJJejZhQkhHcOFfSnBlS2pnUlduclkzR1VmUzNabFY4bGFuREp1dXZiRjBOdDhVeFNGMEdsSlV5c3hZN3ljNwtSV1JJUUgxWU5YVW1
Y2UzYVZsS2xjTnF3MHZTQl8yZOd3dHIWNk1nS1RiV2hmMl8yR2JNWEoybkhSbDlIb1UtVk1rRXdwRExnbkVFdkVJN25INVhJbFJWZ3QzbzhGOHY1d291d1FncXJNVlJKcGw4SlFxQTh6TXZSY0tyRDQtZmlkZlU4NnAzcDk3TEo
ZOhJcWVhRnBmNmlEQTUOZGFFNTAyLWJic2lkbHpObWhFWDNCT2VyME9obS1Dd21VZFRldWVtNEO0VzZGcVdvVjhnbTlROWJhOE53Wk9LUUZaVnl1Zml0eEExZ0NUdDNwOG1UUlRsaGNGVnBMWVJFYjlXNOpJYnBEeGVScOV5eHZ
S09JUmhVUm1EUjBPZDVaQW5NNkZPSzh3VUtLbjdlUzBqODFoWkM5djNzaDY4XOgwdOFqRW4yOXJtWm8yNnNnbFNyaE0tTjNMRk5KRm9QOV9tVGJLQXM4MFhnZ3h6WjdUNkxObi15TXhsTTFwTGtoeE1zbFM2UTJDajROQOJyX19
cGNoQ1lneEpRLU9PTmpPdHdzdC1NbXV2eUotMDJfVTh2RXB3Mms2bUYtSGRiekZKZkRrMHpRdnc3azQxaDFPRWx4TWpwSHZmcOI2Zlh5blpuVmZkUXFqdGxROG1sUOladklPLXNqRldkcWVXbEliYXh1SVNGNnRTOFZ3MDh3TUF
RGF5UXBWVml4TORrcWlNRVlHWjZ4X2pyYmp3c3ZkalZFUXhibWhpelVidnRMemZYcUh6Um4OTlFhdU9ZcGlYSlItbOhrUVhUczJqZHIOOTFFU21fZm52WkNpdUd2V3BzcW9KMFN4LWdHQWJURTE4RFFKWnpLMS1FNVlzaGpjY1F
QlkzTUQOU3ZRU2lpV1Z3Zj3Gc1V6dHU1MONRYXRhTOpkakw4NGVSUEdYYnVfcTJrbnNRQWJ4TFc1dWlTYyO1YnRqbXNQaE4wcFFXNnFTVEZ2dOE3QUdnbDF2ZnlRdlAzeGt0MXNaa3F3cGliUEU3cDVvbDlYVzVzdk85WFYyMGF
UHkxVUhRTGowRVBKeVk3cDBiLUd3VOx5bElMY2F3Lmd6bDNNODRYNO9xdGNmRk1tLTFkaXcmc3RhdGU9MjkOODU5MmItNDRkYyOOOTllLWJiNjUtZGEwMzlhODliMWZh

Updated profile default to use arn:aws:iam::501794745645:root credentials.
[root@ip-172-31-16-171 tmp]# ./create_vpc_user.sh
Creating IAM user: vpc-admin-user...
{
    "User": {
        "Path": "/",
        "UserName": "vpc-admin-user",
        "UserId": "AIDAXJVKMFUWRW4PHQTAN",
        "Arn": "arn:aws:iam::501794745645:user/vpc-admin-user",
        "CreateDate": "2026-11-05T10:53:34+00:00"
    }
}
Attaching AmazonVPCFullAccess policy to vpc-admin-user...
IAM user 'vpc-admin-user' has been created with full VPC access.
[root@ip-172-31-16-171 tmp]#
```

IAM users (3) Info

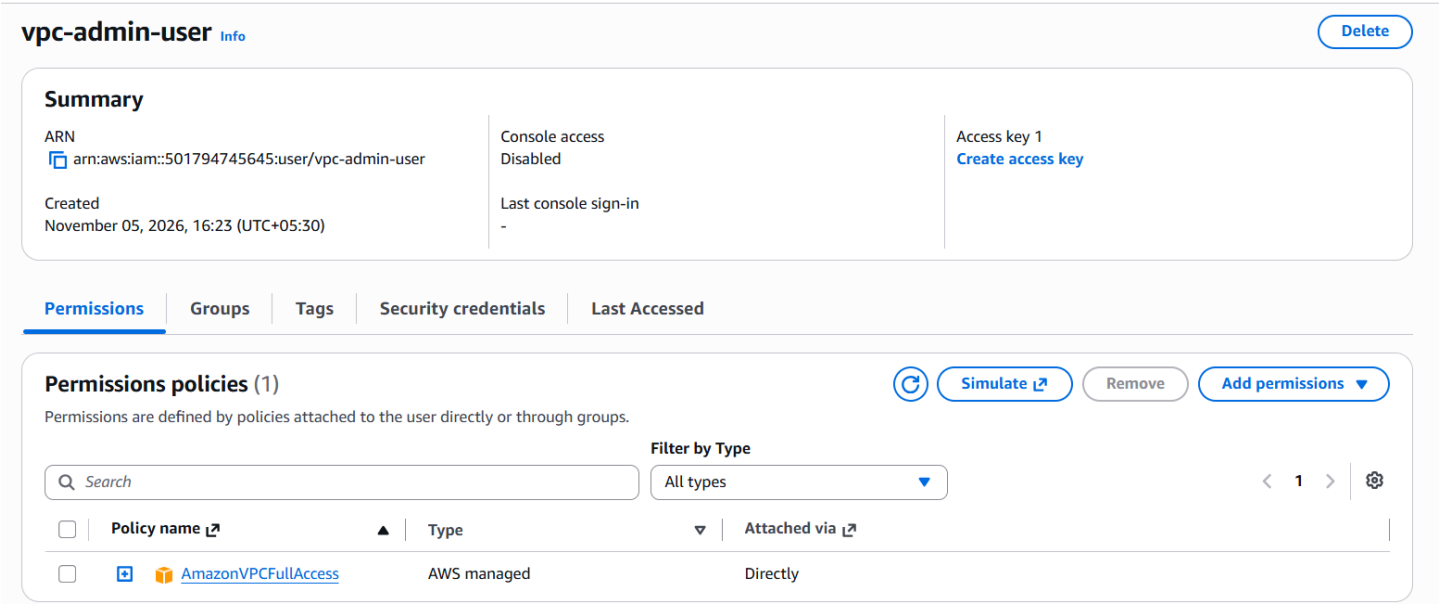
Delete user | Create user

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

1

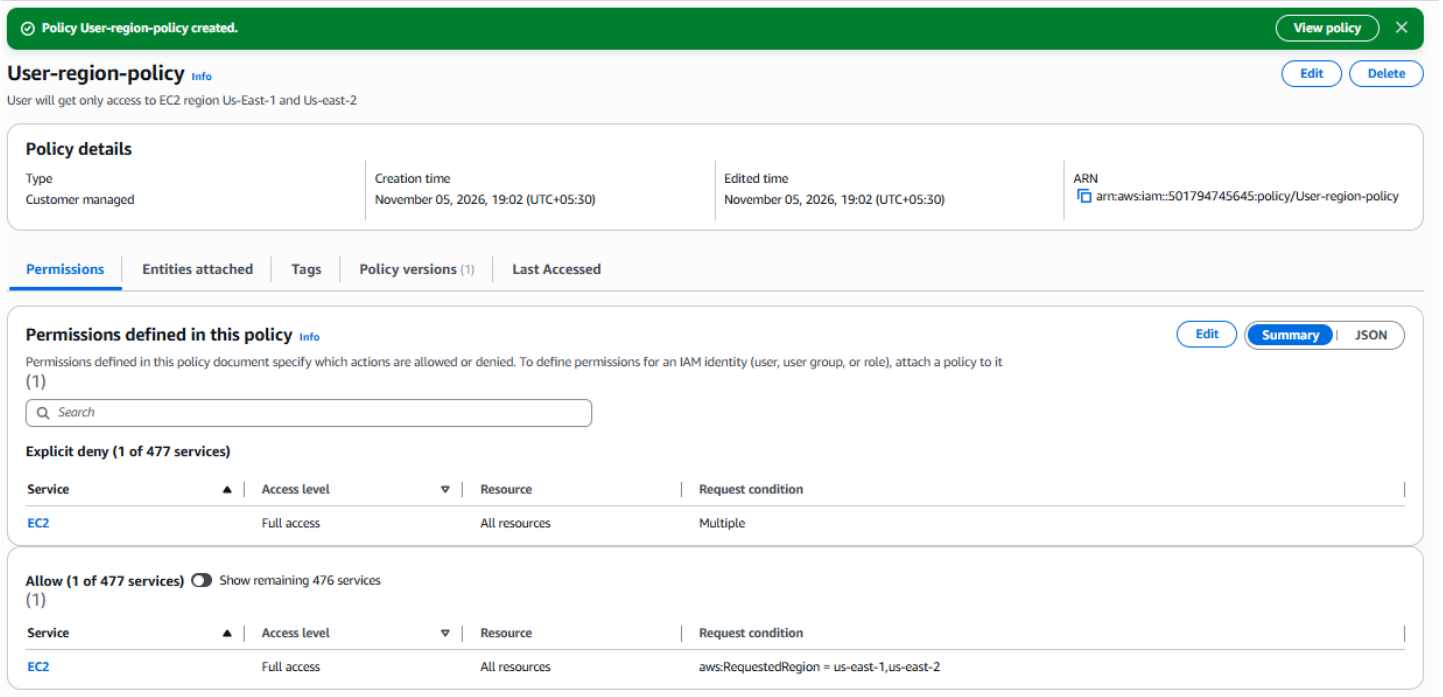
User name	Path	Groups	Last activity	MFA	Password age
Devops	/	1	-	-	-
user-1	/	0	-	-	-
vpc-admin-user	/	0	-	-	-



5. Create an IAM policy to allow EC2 access for a specific user in specific regions only.

Steps:

Go to Policies then created a new policy Region Gave JSON format text code.
then came to USER and Attached that policy.



Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. Learn more

Permissions options

- ○ Add user to group
  Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- ○ Copy permissions
  Copy all group memberships, attached managed policies, and inline policies from an existing user.
- ◉ Attach policies directly
  Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1597)

Choose one or more policies to attach to your new user.

Create policy

Filter by Type

Search | Customer managed | 1 match | 1

Policy name	Type	Attached entities
User-region-policy	Customer managed	0

▸ Set permissions boundary - *optional*

Cancel | Previous | Next

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

EC2-access-user Info

Delete

Summary

ARN arn:aws:iam::501794745645:user/EC2-access-user	Console access Disabled	Access key 1 Create access key
Created November 05, 2026, 19:04 (UTC+05:30)	Last console sign-in -	

Permissions | Groups | Tags | Security credentials | Last Accessed

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Simulate | Remove | Add permissions

Filter by Type

Search | All types | 1

Policy name	Type	Attached via
User-region-policy	Customer managed	Directly

▸ **Permissions boundary** (not set)

▾ **Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn more

Generate policy

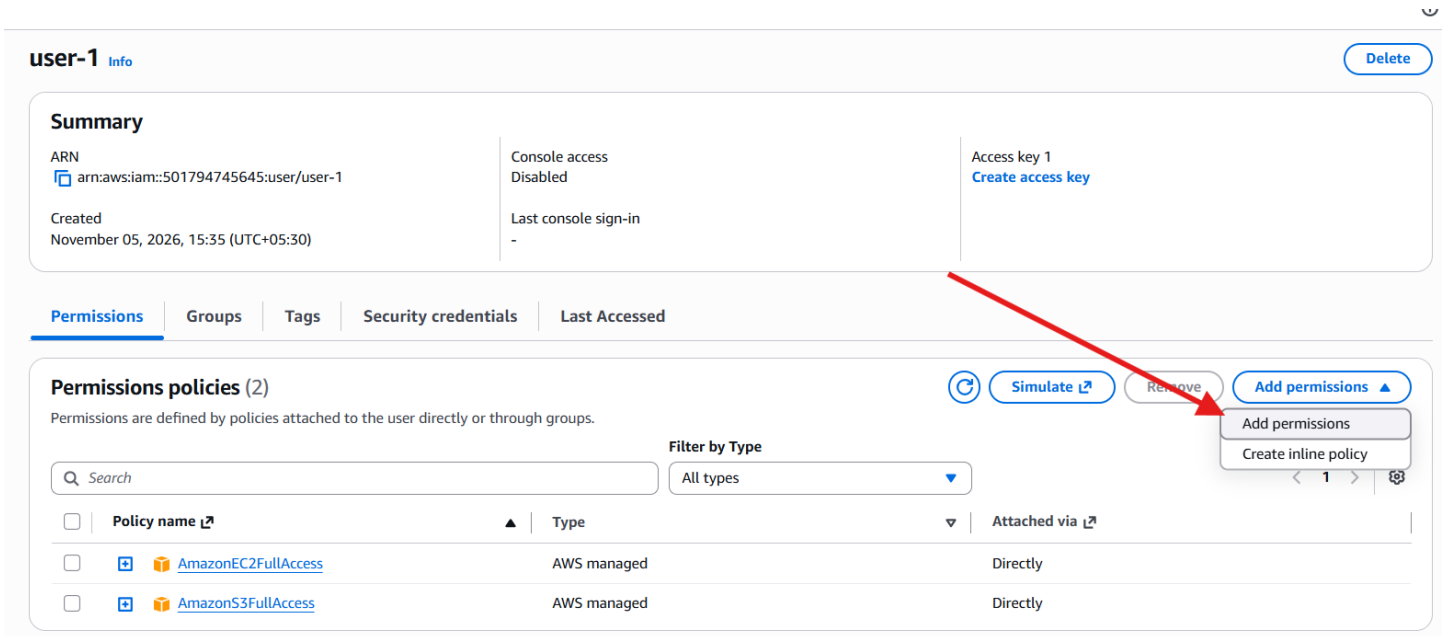
No requests to generate a policy in the past 7 days.

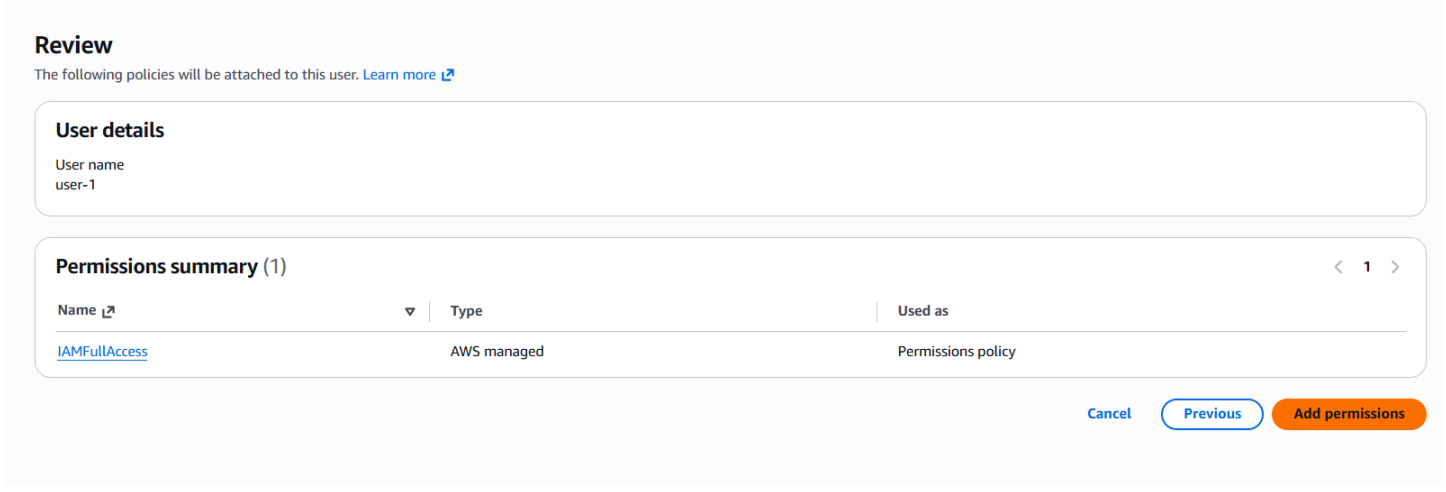
6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.

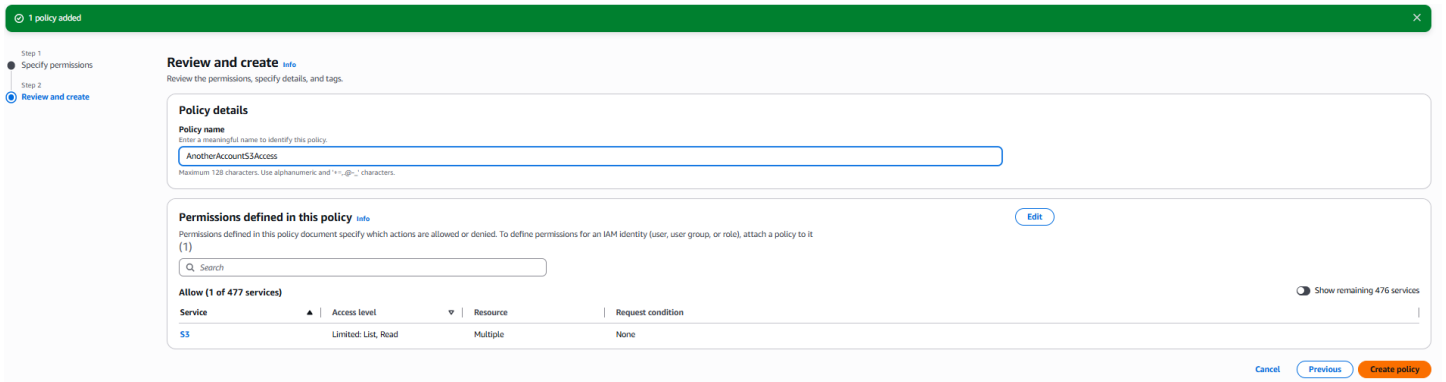
Steps:
Created a user Inline Policy →IAMFullAccess to Access S3 Bucket.
Created S3Bucket disabled the all public access then created a bucket policy.

configured IAM in AWS CLI the try to fetch S3 Bucket from account B.

Results Below:







faisal-s3bucket Info

Objects | Metadata | Properties | **Permissions** | Metrics | Management | File systems | Access Points

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. Learn more about How IAM analyzer findings work

View analyzer for ap-southeast-2

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 bu only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will wo your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

Block *all* public access

⚠ Off

▶ **Individual Block Public Access settings for this bucket**


```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowAccountAListBucket",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::501794745645:user/user-1"
      },
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::faisal-s3bucket"
    },
    {
      "Sid": "AllowAccountAReadObjects",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::501794745645:user/user-1"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::faisal-s3bucket/*"
```

```
Last login: Thu Nov  5 13:55:00 2026 from 103.203.172.230
[ec2-user@ip-172-31-16-171 ~]$ aws --version
aws-cli/2.38.0 Python/3.9.25 Linux/6.18.56-125.164.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-172-31-16-171 ~]$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIAXJVKMFUWQJWRJSA4
AWS Secret Access Key [None]: 2VBFoALPta4vZYMLcb8YEe1UITxXISpiNzrhtGDG
Default region name [None]:
[ec2-user@ip-172-31-16-171 ~]$ aws sts get-caller-identity
{
    "UserId": "AIDAXJVKMFUW3UBHTBCKE",
    "Account": "501794745645",
    "Arn": "arn:aws:iam::501794745645:user/user-1"
}
[ec2-user@ip-172-31-16-171 ~]$ aws s3api head-bucket --bucket faisal-s3bucket
{
    "BucketArn": "arn:aws:s3:::faisal-s3bucket",
    "BucketRegion": "ap-southeast-2",
    "AccessPointAlias": false
}
[ec2-user@ip-172-31-16-171 ~]$ aws s3api head-bucket --bucket faisal-s3bucket
{
    "BucketArn": "arn:aws:s3:::faisal-s3bucket",
    "BucketRegion": "ap-southeast-2",
    "AccessPointAlias": false
}
[ec2-user@ip-172-31-16-171 ~]$ aws s3 ls s3://faisal-s3bucket
2026-11-05 14:16:34      26330 UAN.docx
[ec2-user@ip-172-31-16-171 ~]$
```